# Project Support Session - ReneWind

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# Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

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# ReneWind - Business Context

Renewable energy sources play an increasingly important role in the global energy mix, as the effort to reduce the environmental impact of energy production increases. Out of all the renewable energy alternatives, wind energy is one of the most developed technologies worldwide. The U.S. Department of Energy has put together a guide to achieving operational efficiency using predictive maintenance practices.

Predictive maintenance uses sensor information and analysis methods to measure and predict degradation and future component capability. The idea behind predictive maintenance is that failure patterns are predictable and if component failure can be predicted accurately and the component is replaced before it fails, the costs of operation and maintenance will be much lower.

The sensors fitted across different machines involved in the process of energy generation collect data related to various environmental factors (temperature, humidity, wind speed, etc.) and additional features related to various parts of the wind turbine (gearbox, tower, blades, break, etc.).

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# ReneWind - Business Context

"ReneWind" is a company working on improving the machinery/processes involved in the production of wind energy using machine learning and has collected data on generator failure of wind turbines using sensors. They have shared a ciphered version of the data, as the data collected through sensors is confidential (the type of data collected varies with companies). Data has 40 predictors, 20000 observations in the training set, and 5000 in the test set.

The objective is to build various classification models, tune them, and find the best one that will help identify failures so that the generators can be repaired before failing/breaking to reduce the overall maintenance cost.

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# ReneWind - Data Dictionary

- The data provided is a transformed version of the original data which was collected using sensors.
- Both datasets consist of 40 predictor variables and 1 target variable.

| | V1 | V2 | V3 | V4 | V5 | V6 | V7 | V8 | V9 | V10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | -4.464606 | -4.679129 | 3.101546 | 0.506130 | -0.221083 | -2.032511 | -2.910870 | 0.050714 | -1.522351 | 3.761892 |
| **1** | 3.365912 | 3.653381 | 0.909671 | -1.367528 | 0.332016 | 2.358938 | 0.732600 | -4.332135 | 0.565695 | -0.101080 |
| **2** | -3.831843 | -5.824444 | 0.634031 | -2.418815 | -1.773827 | 1.016824 | -2.098941 | -3.173204 | -2.081860 | 5.392621 |
| **3** | 1.618098 | 1.888342 | 7.046143 | -1.147285 | 0.083080 | -1.529780 | 0.207309 | -2.493629 | 0.344926 | 2.118578 |
| **4** | -0.111440 | 3.872488 | -3.758361 | -2.982897 | 3.792714 | 0.544960 | 0.205433 | 4.848994 | -1.854920 | -6.220023 |

5 rows × 41 columns

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# Open Q&A

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# ReneWind - Evaluation Rubric

| Section | Points |
|---|---|
| Exploratory Data Analysis | 5 |
| Data Preprocessing | 4 |
| Model Building | 6 |
| Model Performance Improvement and Final Model Selection | 34 |
| Actionable Insights & Recommendations | 3 |
| Presentation/Notebook - Overall Quality | 8 |

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# ReneWind - Evaluation Rubric

## **Section 1**: Exploratory Data Analysis

| What to Cover? | Guiding Questions and Considerations |
| --- | --- |
| Data Overview | Get a glance of the data and attributes |
| Univariate Analysis | Do multiple attributes have a similar distribution? |
| Bivariate Analysis | How do predictor variables vary with the target variable? |

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# ReneWind - Evaluation Rubric

## **Section 2**: Data Preprocessing

| What to Cover? | Guiding Questions and Considerations |
| --- | --- |
| Prepare the data for modelling | Is the ratio of the dataset split into training and testing sets appropriate? |
| Missing Value Treatment | Is the rationale for missing value treatment logical?<br>Is data leakage taken care of? |

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# ReneWind - Evaluation Rubric

## **Section 3**: Model Building

| What to Cover? | Guiding Questions and Considerations |
| --- | --- |
| Choose the metric of choice with rationale | Business end goal drives the evaluation metric of choice |
| Build a Neural Network with SGD as the optimizer | Should I expect this to be best performing model? |
| Comment on model performance | Focus on the metric of choice, but be mindful of the cons too |

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# ReneWind - Evaluation Rubric

**Section 4**: Model Performance Improvement and Final Model Selection

| What to Cover? | Guiding Questions and Considerations |
| --- | --- |
| Use combinations (at least 6) of different methods (More hidden layers, Different optimizers, Dropout, Class Weights) to improve the model performance | How many more hidden layers do I add?<br>Some optimizers converge faster than others<br>Is the class distribution imbalance enough to use class weights? |
| Comment on model performance for each of the models | What's the effect of different strategies explored? |
| Choose the best model from the ones built with proper reasoning | Business end goal drives the best model choice |

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# ReneWind - Evaluation Rubric

## **Section 5**: Actionable Insights and Recommendations

| What to Cover? | Guiding Questions and Considerations |
| --- | --- |
| Conclude with the key takeaways for the business | What are the key things that we learned from the data that's useful for the business?<br><br>Logical next steps are a key inclusion |

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# ReneWind - Evaluation Rubric

**Section 6**: Presentation/Notebook - Overall Quality

| Low-code Considerations | Full-code Considerations |
| --- | --- |
| Clear structure, flow, and visual appeal - everything sits well in a story | Clear structure and flow |
| Crispness - Focus on key points | Well-commented and executed code |
| All sections of the rubric included, even if in an appendix | All sections of the rubric included, even if in an appendix |

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# ReneWind - Low-Code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

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# ReneWind - Low-Code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

**The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file**

**Please make sure that all the sections mentioned in the grading rubric have been covered in the submission**

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# ReneWind - Full-Code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

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# ReneWind - Full-Code Version

**The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)**

The conversion can be done via one of the following ways:

**Jupyter Notebook**: ***Download as HTML*** from the ***File*** menu
**Google Colab**: Use free online tools

**Please make sure that all the sections mentioned in the grading rubric have been covered in the submission**

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# ReneWind - FAQ

**I keep getting the following error while installing TensorFlow:**

```
ImportError: DLL load failed while importing _pywrap_tensorflow_internal: The specified module could not be found. Failed to load the native TensorFlow runtime.
```

If you want to use the Tensorflow Version 2 with the least dependencies, use the below command:

```
!pip install tensorflow==2.15.0
```

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# ReneWind - FAQ

**I keep getting the below error while executing the fit function for the neural network. Why?**

```
ValueError:  Input 0 of layer sequential_55 is incompatible with the layer: expected axis
-1 of input shape to have value 10000 but received input with shape (700, 13)
```

This error is due to the incompatible shape of the input layer. Please pass the input shape in the first layer as follows:

```
v = X_train.shape
s = v[1]
model2.add(Dense(number_of_units, input_shape=s, kernel_initializer='he_normal',
activation='relu'))
```

This should use a compatible shape for the input layer that prevents the error from occurring.

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# Power Ahead!

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